

ChinookAnalytics: SQL Analytical Data Platform

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February 18, 2026

Abstract

ChinookAnalytics is a SQL-based analytical data platform designed to transform raw transactional music-store data into validated, structured, and business-aligned analytical assets.

The project implements a full layered pipeline (staging, core, marts), enforces relational integrity, defines formal analytical objectives, and produces decision-oriented outputs supported by reproducible validation.

This document presents the architectural design, modeling strategy, analytical methodology, and summary findings of the platform.

1 Product Scope and Objectives

ChinookAnalytics is designed as an end-to-end analytical engineering system. Its primary objectives are:

- Ingest raw relational data in a controlled environment,
- Enforce referential and domain integrity at the core layer,
- Translate business questions into formal analytical models,
- Materialize reproducible analytical marts,
- Validate aggregation consistency across layers.

The system treats analytics as a structured product rather than an exploratory exercise.

2 Source Dataset: Chinook

The Chinook dataset represents a simplified digital music retail operation where customers purchase individual tracks through invoices.

Core structural characteristics:

- 412 invoices
- 2240 invoice lines
- 59 customers

- 275 artists
- 3503 tracks
- 24 billing countries
- 60 months of transactions (2021–2025)

Revenue is defined at invoice level via `invoice.total`, with line-level decomposition through `unit_price` and `quantity`.

3 Layered Architecture

The project follows a strict three-layer architecture.

3.1 Staging Layer (stg)

Raw CSV files are ingested without transformation. This layer preserves source fidelity.

3.2 Core Layer

The core schema enforces:

- Primary keys,
- Foreign keys,
- Domain checks (non-negative monetary values),
- Referential completeness validation.

Profiling confirmed:

No orphan invoices
 No orphan invoice lines
 No negative totals
 No missing primary keys

This guarantees structural correctness before analytics.

3.3 Marts Layer

The marts layer materializes analytical models aligned with formally defined business objectives.

Implemented marts:

- `mart_revenue_time`
- `mart_market_performance`
- `mart_customer_economics`

- mart_portfolio_artist
- mart_portfolio_genre
- mart_revenue_composition
- mart_dependency_summary

Each mart has a defined grain and metric specification.

4 Analytical Framework

Three formal documents define the analytical structure of the project:

- **ChinookAnalytics_data_discovery.pdf** Structural and statistical profiling of the curated core schema.
- **ChinookAnalytics_analysis_objectives.pdf** Formal mapping of stakeholder questions into measurable metrics and mart definitions.
- **ChinookAnalytics_final_report.pdf** Business interpretation derived from the materialized marts.

These documents are integral components of the platform and define its analytical contract.

5 Key Structural Findings (Data Discovery)

The discovery phase identified:

- Stable invoice totals (min 0.99, max 25.86),
- Median invoice value: 3.96,
- P90 invoice value: 13.86,
- Quantity fixed at 1 across all invoice lines,
- Balanced yearly revenue (449–481 range).

Limitations identified:

- No discount or promotion variable,
- No basket scaling (quantity constant),
- Limited behavioral variation for churn modeling.

6 Analytical Results Summary

6.1 Revenue Stability

- Total revenue: 2328.60
- Revenue consistent across five years
- No significant upward or downward trend

Revenue demonstrates structural stability.

6.2 Market Concentration

- Top 5 countries: 58.78% of revenue
- Top 10 countries: 76.05% of revenue

Moderate geographic concentration risk exists.

6.3 Customer Concentration

- Top 10 customers: 19.38% of revenue
- Top 20 customers: 36.61% of revenue

Customer base is moderately diversified.

6.4 Artist and Genre Distribution

- Top artist share: 5.95%
- Top 10 artists: 30.98%
- Rock genre largest contributor

No single entity dominates the revenue structure.

7 Validation Framework

A reconciliation script verifies aggregation integrity.

```
core_rev      = 2328.60
time_rev      = 2328.60
market_rev    = 2328.60
customer_rev  = 2328.60
artist_rev    = 2328.60
genre_rev     = 2328.60
comp_rev     = 2328.60
dep_rev       = 2328.60
```

All analytical layers reconcile exactly with core totals.

8 Operational Environment

The system is deployed using Docker Compose:

- PostgreSQL container
- Layered SQL execution scripts
- Deterministic pipeline execution
- Full reproducibility

Pipeline order:

`stg → core → profiling → marts → validation`

9 Conclusion

ChinookAnalytics delivers a structured analytical data platform built entirely in SQL.

The system:

- Enforces structural integrity,
- Formalizes analytical objectives,
- Produces validated analytical marts,
- Generates business-aligned insights.

The project demonstrates end-to-end analytical engineering, from ingestion to strategic interpretation, within a reproducible relational environment.